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SHORT COMMUNICATION

Serological Survey of Hantavirus and Flavivirus Among Wild Rodents in Central Italy

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Abstract

Hantaviruses are a group of zoonotic viruses carried by rodents. Puumala virus (PUUV) and Dobrava virus (DOBV) are the causative agents of human hantavirus infections in Europe. Knowledge about hantavirus circulation in Italy is very scarce. West Nile virus (WNV) and Usutu virus (USUV) are emerging neuropathogenic flaviviruses, both endemic in most part of the Italian territories. To monitor the circulation of PUUV, DOBV, WNV, and USUV in natural environment in central Italy, we carried out serological surveillance in wild rodents. During this study, 90 animals were captured in forested areas of Abruzzo and Marche regions and tested with serological assays for the specific pathogens. Serological test provided no evidence of PUUV and DOBV circulation in the studied area. However, four rodents (*Apodemus flavicollis*) were found to be positive by WNV ELISA test. Two of them were confirmed as WNV by virus neutralization test.

Keywords: flavivirus, hantavirus, Italy, serological survey, wild rodents

Introduction

Hantaviruses, members of the family Bunyaviridae, are rodent-borne viruses causing the human disease known as hemorrhagic fever with renal syndrome (HFRS). Hantaviruses are enveloped RNA viruses that have rodents and insectivores as hosts and can be transmitted to humans by aerosols from host excreta or by direct contact.

The most prominent and widely occurring hantavirus in Europe is Puumala virus (PUUV), transmitted by the bank vole (*Myodes glareolus*). PUUV causes a mild form of HFRS, called nephropathia epidemica. Dobrava virus (DOBV) is transmitted by the yellow-necked field mouse (*Apodemus flavicollis*) and is known to cause more severe forms of HFRS (ECDC 2016). No cases of human hantavirus have been reported in Italy, although limited information is available on the circulation of these viruses in the country (Kallio-Kokko et al. 2006, ECDC 2016).

West Nile virus (WNV) and Usutu virus (USUV) are emerging neuropathogenic flaviviruses. Both viruses are maintained in the environment through a bird–mosquito life cycle (Hubálek and Halouzka 1999). Spillover from this cycle occasionally occurs and causes disease in mammalian hosts, in particular in humans and horses. Exposures to WNV and USUV have been documented in a variety of wild mammals (Jeffrey 2013). Association of WNV and USUV with wild rodents has been observed in field and experimental studies. In North America, certain species of tree squirrels (*Sciurus* spp.) and eastern chipmunks (*Tamias striatus*) were found to develop viremia sufficient for infecting mosquitoes after natural infection (Jeffrey 2013). Furthermore, experimental infections showed that certain WNV lineages are highly pathogenic for laboratory mice (Pérez-Ramírez et al. 2017), thus highlighting the potential WNV has for infection in rodents.

WNV appeared for the first time in Italy in 1998 causing encephalitis in horses in Tuscany. Ten years later, WNV reappeared in northern Italy affecting horses and humans (Monaco et al. 2010). To monitor and control WNV circulation, a surveillance program has been implemented at the national level by the Italian Ministry of Health (CESME, 2016). Within this framework, WNV was detected in birds, mosquitoes, and equids (Monaco et al. 2010), and several human cases were reported (Rizzo et al. 2016). USUV circulation was demonstrated in Italy since 1996. In the last decade, the virus has been associated to two cases of human encephalitis in northern Italy, confirming the zoonotic potential of this virus (Pecorari et al. 2009).

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The aim of this study was to monitor the circulation of hantavirus, WNV, and USUV in wild rodents in areas of central Italy.

Material and Methods

The study was carried out according to the Italian (Decreto legislativo 04.03.2014 n°26) and European (Directive 2010/63/UE) regulation for Animal Welfare. Formal authorization has been provided by the Interuniversity Ethics Committee for Animal Experimentation (CEISA) on October 29, 2014 (ID no.: 1466/2014).

Wild rodents were trapped in 24 sites of Abruzzo and Marche Regions using Sherman live traps, from July 2015 to April 2016 (Supplementary Fig. S1; Supplementary Data are available online at www.liebertpub.com/vbz). The study areas were selected at altitudes not exceeding 1000 m with the presence of mixed forest according to ecological and environmental conditions favorable to wild rodents. Mosquitoes and ticks, potential vectors of flaviviruses, were abundant as observed during the field activity. Rodent species were determined morphologically. Once captured, the animals were inspected for determining sex, age, and a small amount of blood was collected from the mandibular plexus for serological testing. Once the data registration and the blood collection were completed, the animals were released in the capture sites. Serological testing for PUUV was carried out using a PUUV-IgG immunochromatography test, Reascan_ Ab-Dect PUUMALA IgG (Oy Reagena Ltd., Toivala, Finland), according to the manufacturer's instructions. Diagnosis of DOBV was carried out using the ELISA Kit Ab-Dect Dobrava–Hantaan IgG EIA (Oy Reagena Ltd.). The prevalence levels of PUUV in *Apodemus* spp. and DOBV in *Myodes* were analyzed using Bayesian approach with a Beta distribution (Sivia 1996) allowing at reporting the 95% confidence interval (CI). Serum samples were tested with the competitive ELISA INgezim WNV Compac (Ingenasa, Madrid, Spain), which crossreacts with different flaviviruses. An approach based on detecting WNV-crossreacting antibody is useful for surveillance purpose, but may affect the diagnostic accuracy, mainly for USUV. Therefore, the results obtained were interpreted with caution and the positive and doubtful ELISA results were further analyzed in parallel with the virus neutralization (VN) tests (microtitre format) for WNV and USUV (Di Gennaro et al. 2014).

Results

Ninety wild rodents: *A. flavicollis* (60), *Apodemus sylvaticus* (6), *M. glareolus* (22), and *Eliomys quercinus* (2) were trapped in forested areas of Abruzzo and Marche regions (Supplementary Fig. S1) and tested with serological assays specific for PUUV, DOBV, WNV, and USUV. The diagnosis of PUUV was performed directly on-field and no positive reactions were observed out of 90 samples analyzed. Then sera were tested with the competitive ELISA INgezim WNV Compac for WNV giving four positive and two doubt results out of 90 sera tested. Only two out of four sera, positives in ELISA, were confirmed as WNV by VN test, with neutralizing antibody titers of 1:10. None was positive for USUV. The WNV confirmed cases were both *A. flavicollis* captured in Abruzzo in 2015 and in Marche region in 2016, respectively. Forty-six sera from *A. flavicollis* (43) and *A. sylvaticus* (3) were tested for DOBV giving any positive result (Table 1). The estimated prevalence of hantaviruses was 0 (95% CI: 0–12, 21%) for PUUV in *M. glareolus* and 0 (95% CI: 0–6, 18%) for DOBV in *Apodemus* spp.

Discussion

Hantavirus infections are widely distributed across Europe. More than 3700 human cases of hantavirus were reported in the European countries in 2014, most of them in Croatia, Germany, France, Finland, and Sweden (ECDC 2016). In Mediterranean countries a low number of cases are usually reported although reservoir host are present (ECDC 2016). Landscape ecology, rodent population densities, and seroprevalence levels are known to correlate with human cases of disease. In Italy no human cases have been reported so far. A serosurvey on wild rodents carried out in 2000–2003 in the north eastern region of Trentino indicated low seroprevalences for PUUV in *M. glareolus* (0.4%) and for DOBV in *A. flavicollis* (0.2%) (Kallio-Kokko et al. 2006). Our results suggested absence of hantavirus circulation in the areas of Central Italy, where the study has been realized, or if these pathogens are present their prevalence levels, with 95% probability, are lower than the CIs estimated.

Since the reintroduction in 2008, a constant and intensified circulation of WNV was observed across Italy. From 2008 to 2015, WNV was detected in mosquitoes, birds, and horses in many of the 21 regions of Italy (Rizzo et al. 2016). According to the national plan for WNV surveillance, the regions of Abruzzo and Marche are considered nonendemics. In our study, we detected two WNV-positive *Apodemus* captured in

Table 1. The Rodent Species, Number of Animals Trapped, and Tested and the Positive Results for Hantavirus and Flavivirus Serological Assays

	Hantavirus				*Flavivirus*			
	PUUV		*DOBV*		*WNV (ELISA)*		*WNV (VNT)*	*USUV (VNT)*
Rodent species	*Tested*	*Pos*	*Tested*	*Pos*	*Tested*	*Pos*	*Pos/tested*	*Pos/tested*
Apodemus flavicollis	60	0	43	0	60	4	2/4	0/4
Apodemus sylvaticus	6	0	3	0	6	0	—	—
Myodes glareolus	22	0	—	—	22	0	—	—
Eliomys quercinus	2	0	—	—	2	0	—	—

DOBV, Dobrava virus; PUUV, Puumala virus; USUV, Usutu virus; VNT, virus neutralization test; WNV, West Nile virus.

forested sites of Abruzzo and Marche, respectively. This finding together with the lack of any previous detection of WNV in birds or horses in these regions may indicate low level of WNV circulation limited to the forested areas. However, serological crossreactions with related flavivirus that may be circulating in the same areas could not be excluded when considering the low virus neutralization titers observed in the two WNV-VNT-positive rodents and the presence of ELISA-positive *Apodemus*, which reacted negatively with the WNV and USUV VN tests. This study pinpoints the susceptibility of *Apodemus* to flavivirus, confirming observations reported in other studies (Hubálek and Halouzka 1999); however, the role of this species in the maintenance and spread of flavivirus in the environment together with their potential use as sentinels in their habitat warrant further investigation.

Conclusions

Active surveillance of wild rodents provided important insight on the circulation of hantavirus and flavivirus in Central Italy, allowing more precise assessment of the public health risk associated to these pathogens. This activity, therefore, should be improved on a broader scale, eventually including other pathogens relevant for the human health.

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Author Disclosure Statement

No competing financial interests exist.

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